Exploratory Data Analysis of Retail Sales Data

Mission DS01 · Week 1 — Introduction to Data Science with Python

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Data period: 1 January – 31 December 2025

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How to read this report. Every number comes from the notebook *Week1_EDA.ipynb*. **Revenue means Completed orders** unless a figure says 'all orders'. Amounts are in Nigerian naira (₦), an assumption, because the data does not state a currency.

1. Executive Summary

ShopSmart supplied nine linked tables covering **5,000 customers, 500 products, 25 stores and 50,000 orders** placed in 2025. We loaded, cleaned and merged the data, calculated the key performance indicators, created ten required charts, studied correlations, engineered new features and answered ten business questions. In short: **sales are steady and evenly spread, but only a third of orders are completed and several parts of the data fail basic logic checks.**

₦2.50bn Revenue (Completed orders)	**16,529** Completed orders (of 50,000)	**₦151,057** Average order value	**4.88%** Return rate (all orders)
33.0% of booked value is Completed	**24.3%** of orders sold below cost	**₦3.17bn** Inventory value at cost	**23 / 307** products to restock / overstocked

What we found

- **Only one third of booked value is Completed.** Of ₦7.58bn booked, ₦2.50bn (33.0%) is Completed, ₦2.53bn is Pending and ₦2.55bn (33.7%) was Cancelled.
- **Performance is very even.** Stores, regions, customer segments, categories and months are alike; the store ranking even changes depending on whether all orders or only Completed orders are counted.
- **Pricing data is unreliable.** 24% of orders were sold below cost, the price charged is unrelated to the price list ($r \approx 0$), and the discount is subtracted as a flat amount rather than a percentage.
- **Return records fail logic checks.** Half of the returns are dated before the order, two-thirds belong to orders that were never completed, and 270 refunds exceed the sale.
- **Inventory is unbalanced.** 23 products need restocking, yet 307 products hold more than a year of stock worth ₦2.72bn (86% of inventory value).

Bottom line. Fix order fulfilment and data capture first. The data supports reporting, anomaly detection and rule-based inventory control today, but not reliable prediction.

2. Dataset Description

ShopSmart supplied nine CSV tables. File names differ slightly from the brief (*ShopSmart_*.csv*; the customer file starts with *ALA_*).

Table	Rows	Columns	One row per…	Key fields
Customers	5,000	11	customer	Age, State, Segment, Registration_Date
Products	500	7	product	Category, Brand, Cost_Price, Selling_Price
Orders	50,000	12	order (one product each)	Customer, Product, Store, Date, Quantity, Unit_Price, Discount, Status
Order_Items	50,000	7	order	Line_Total (otherwise a copy of Orders)
Stores	25	6	store	Region, State, City, Opening_Date
Returns	2,500	5	return record	Order_ID, Return_Date, Reason, Refund_Amount
Employees	125	6	employee	Store_ID, Role, Hire_Date, Monthly_Salary
Suppliers	489	6	supplier	Supplier_Name, Country
Inventory	500	6	product (in one store)	Stock_On_Hand, Reorder_Level

Points to know about the data

- **Tables link by ID** (customer, product, store, order). Suppliers link to products only through the brand name (supplier name = brand + ' Supplies').
- **Orders and Order_Items are copies.** Every order appears once in each table with identical fields; Order_Items only adds Line_Total.
- **Inventory has one row per product** (each product is held in a single store), so it must never be merged onto the 50,000 orders.
- **One year of data** (2025), so trends and seasonality cannot be separated.
- **No data dictionary detail:** units (currency, the meaning of Discount) had to be inferred and are listed as assumptions in Section 9.
- **Distributions are unusually flat:** quantity, discount, price, shipping, age and the status/channel/payment splits are almost perfectly even. Real retail data is normally uneven, so strong patterns should not be expected.

3. Data Cleaning Process

The rule followed was: **fix what is clearly wrong, flag what is suspicious, never delete.** 39 steps are documented in *Cleaned_Data/Cleaning_Log.csv*; no rows were removed. Suspicious records carry a True/False `flag_` column so they can be included or excluded.

Task (from the brief)	What was done	Result
Remove duplicates	Checked all nine tables for repeated rows and repeated IDs	None found
Handle missing values	Checked every column; looked at the odd phone-number formats	No missing values; phones left as supplied (not used)
Fix data types	Labels (status, channel, payment, category) stored as category; customer segment made ordered (Bronze < Silver < Gold)	Smaller, clearer tables
Convert dates	Six date columns converted with an explicit month/day/year format	Dates run 1 Jan – 31 Dec 2025
Rename columns	All names lower-case; repeated names given prefixes (customer_state, store_state, supplier_country…)	Merges stay readable
Check inconsistent values	Logic checks across tables (below)	Several problems flagged

Inconsistencies found (rows kept and flagged)

Problem	Count	Why it matters
Orders and Order_Items are identical copies	50,000	Only Line_Total is brought across; avoids double counting
Line_Total subtracts Discount (0–20) as a flat amount	all rows	If Discount is a percentage, revenue is about 10% lower (roughly ₦250m on Completed orders); the client must confirm
Orders sold below the product cost price	12,163 (24.3%)	Negative-margin sales; price data unreliable
Orders dated before the customer registered	5,159	Registration dates unreliable
Returns dated before the order	1,252 (50%)	Impossible; time-to-return cannot be used
Returns on Pending or Cancelled orders	1,685 (67%)	Goods never delivered cannot be returned
Refund larger than the sale	270 (11%)	Refund values unreliable
Order returned twice	114 records	Returns are totalled per order before merging
Store names repeat (e.g. four 'ShopSmart Abuja'); region is not geographic	25 stores	Always group by store ID; regions need re-defining
Gender label does not match first names	5,000	Gender not used in any conclusion

Merging

Separate analysis tables were built at the correct level: a **sales table** (one row per order, joined to products, stores, customers and total refunds), a **customer table** (one row per customer) and an **inventory table** (one row per product). The sales table is checked to contain exactly 50,000 unique orders after merging. Returns were totalled per order first so that orders returned twice are not counted twice.

4. Exploratory Findings

4.1 Key performance indicators

Each KPI states its definition. Revenue, average order value and quantity all use **Completed orders** so that they agree with each other; the return rate uses all orders because returns exist on orders of every status.

KPI	Value	Definition
Total Revenue (Completed orders)	2,496,824,383	Sum of line_total, Completed orders
Net of refunds	2,475,234,402	Revenue minus refunds on Completed orders
For comparison: all orders booked	7,575,250,086	line_total of every order, including Pending and Cancelled
Total Customers	5,000	Unique customers in the customers table
Total Products	500	Unique products in the products table
Total Stores	25	Unique stores in the stores table
Total Orders (all)	50,000	Unique order IDs
Completed orders	16,529	Orders with status Completed
Average Order Value (Completed)	151,057	Completed revenue ÷ completed orders
Average Selling Price (catalogue)	36,244	Mean price-list price over the 500 products
Average Selling Price (charged per unit)	50,510	Completed revenue ÷ units sold
Total Quantity Sold (Completed)	49,432	Units on Completed orders
Return Rate (all orders)	4.88%	Orders with a return ÷ all orders
Return Rate (Completed orders)	4.83%	Completed orders with a return ÷ Completed orders
Inventory Value (at cost)	3,169,020,818	Stock on hand × cost price, from the inventory table

4.2 Correlation analysis

Strongly correlated: line total with unit price (r = 0.73) and with quantity (r = 0.60) — a formula (line total = quantity × price − discount); and selling price with cost price (r = 0.96) — the price list is cost plus a mark-up. **Unrelated:** customer age, shipping cost and discount have |r| below 0.05 with everything else. The price actually charged is unrelated to the catalogue price (r ≈ 0). **Uses:** check prices against the cost-plus pattern; never use a calculated column to predict its own ingredients; accept that these columns cannot predict customer behaviour; test whether discounts really raise sales volume (they do not appear to).

4.3 Answers to the ten business questions

Q1. Which products generate the most revenue?

The top product is Beauty Item 210 (8,075,117), followed by Groceries Item 444 and Groceries Item 304. The top 10 products together give 3.1% of revenue (if all 500 were equal it would be 2%). The best product earns only 1.6 times the typical (median) product, so revenue is spread very evenly. Note: only 1 of these 10 products is also in the top 10 when all orders are counted, so product revenues are very close.

Q2. Which stores perform best?

On Completed orders the best stores are STR003 Lagos, STR007 Ibadan, STR021 Lagos. The best store is +13.6% vs the average store and the weakest (STR002 Kano) is -9.5%. The number of completed orders (about 661 per store) explains most of the difference, not order size. Caution: on ALL orders the best store is STR015 Lagos and the gap is only 6.0%, so no store is clearly better than the others.

Q3. Which customer segments spend the most?

Gold customers spend the most per customer, but the gap between the best and worst loyalty segment is only 0.8% (age groups: 5.7%). The Gold tier does not spend noticeably more than Bronze. The real differences are between individual customers: the top 10% of customers give 23.9% of revenue.

Q4. Which products have high return rates?

The highest return rates are Home Item 263, Beauty Item 262, Beauty Item 443 (up to 11.6%) against an overall rate of 4.88%. But each product has only about 100 orders, so 5 returns is a 5% rate and 9 returns is 9%: a difference of a few returns is normal luck. By category the rates are all close (4.58% to 5.13%). No product is clearly a problem.

Q5. Which months have peak sales?

The highest-revenue months are December, October, January; the lowest is February, partly because it has fewer days. Per day, the best month (December) is 11.8% above the worst. There is no strong seasonal peak in 2025.

Q6. Which categories are growing?

Comparing July-December with January-June, Fashion grew most (+6.4%) and Home changed least (+2.1%). But revenue normally swings by about 10% from one month to the next, so these changes are within normal ups and downs. With only one year of data we cannot say any category is truly growing.

Q7. Which suppliers perform best?

Using revenue, margin and return rate together, the highest-scoring suppliers are Robinson, Baker and Knight, Brown-Williams, Poole-Whitney. Treat this with care: each supplier has only about 102 orders, margin mostly reflects the product's cost price rather than how well the supplier performs, and the four countries are very similar. To judge suppliers properly ShopSmart needs delivery time, order accuracy and defect data.

Q8. Which inventory items require restocking?

23 products need restocking now (21 are at or below their reorder level). The most urgent are PROD0403, PROD0479, PROD0059. Ordering enough for 60 days of sales needs 322 units costing about 7,859,732. The opposite problem is bigger: 307 products (61%) have more than a year of stock, tying up 2,716,869,461.

Q9. Which regions underperform?

By total revenue the West region is lowest, but it has only 4 stores (average 6.2 per region). Per store, the lowest region is West (-2.6% vs the average store) and the highest is North (+1.5%). By state the lowest per store is Kano. These gaps are small, and the region field is unreliable (the same city appears in several regions), so no region clearly underperforms.

Q10. What patterns could support predictive modelling?

Very few patterns in this data could support a predictive model. Return rates are about 5% in every group (channel, payment, category, discount, age, segment), so none of these helps predict returns; sales are flat across months, stores, regions and categories; customer spend is the same across segments and ages. What does exist are structural links (line total follows quantity × price; selling price follows cost price, $r \approx 0.96$). Anomaly detection is feasible now; customer segmentation and rule-based inventory planning are partly feasible; sales forecasting, return prediction, churn and recommendation need better data first.

5. Visualisations

Charts 1–10 are the ten required by the brief; 11–16 support the correlation and business-question sections. All are saved in *Visualizations/*. Bars start at zero so small differences are not exaggerated.

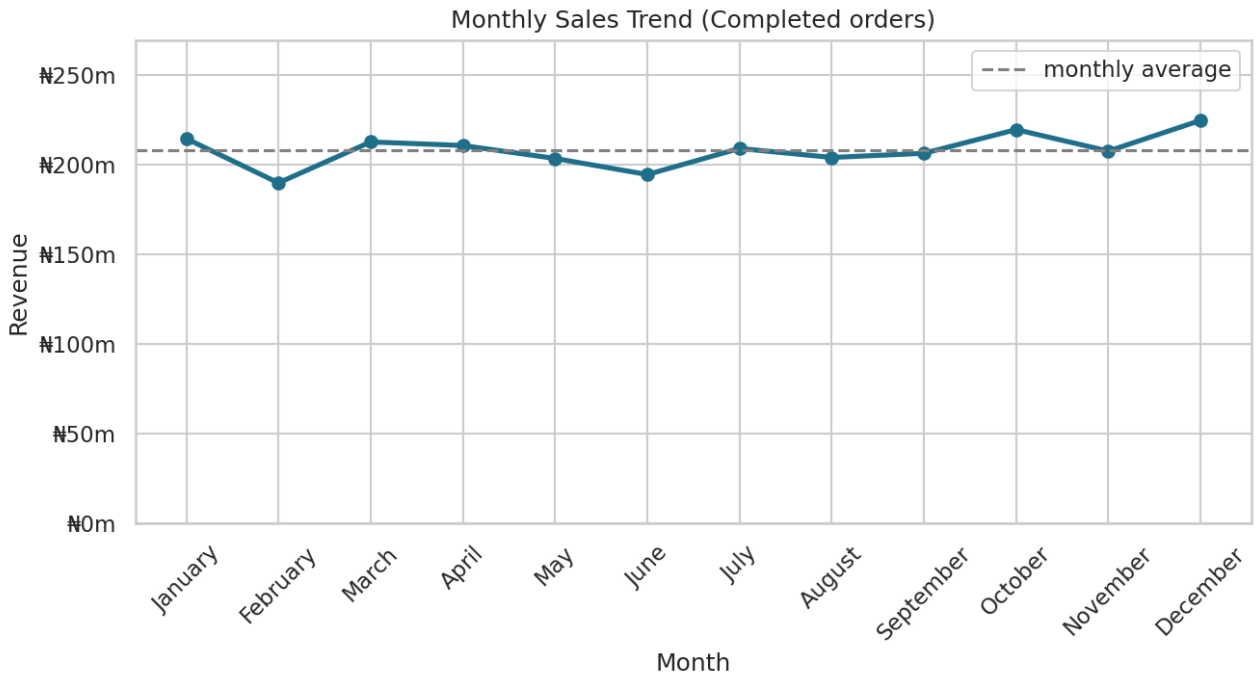


Figure 1 — Monthly sales trend (Completed orders). The line is fairly flat; December is highest and February lowest (a shorter month).

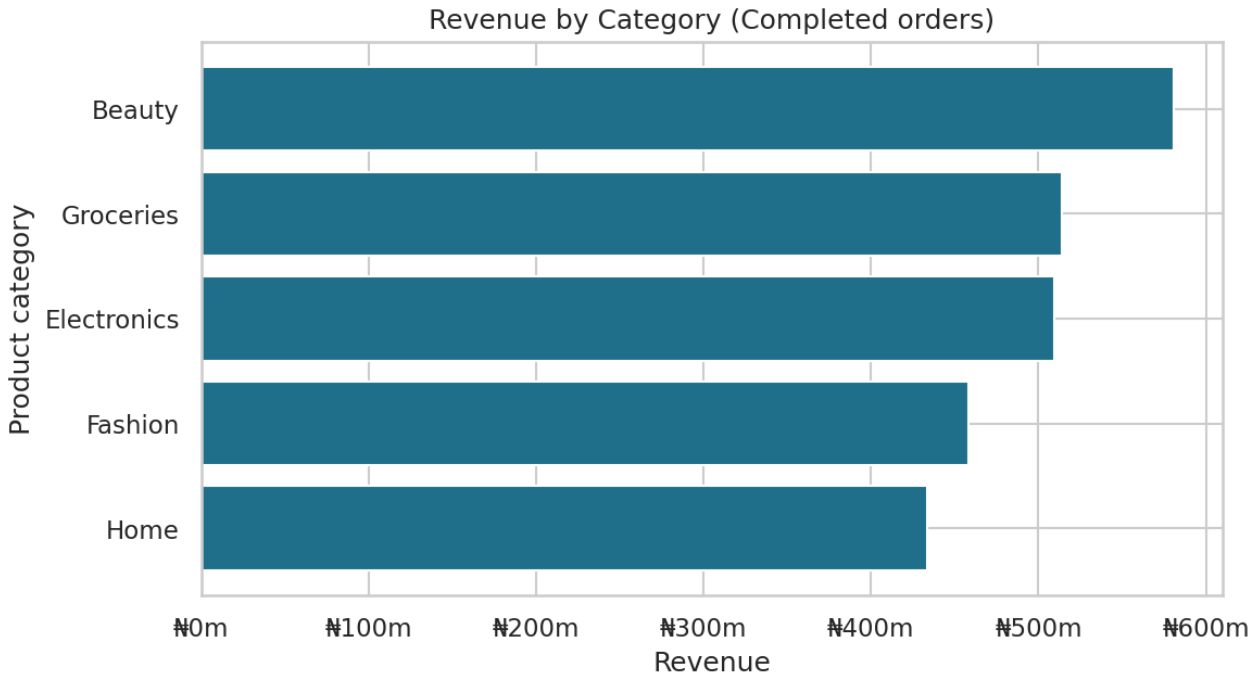


Figure 2 — Revenue by category. Beauty leads because it has the most products (119); revenue per product is similar across categories.

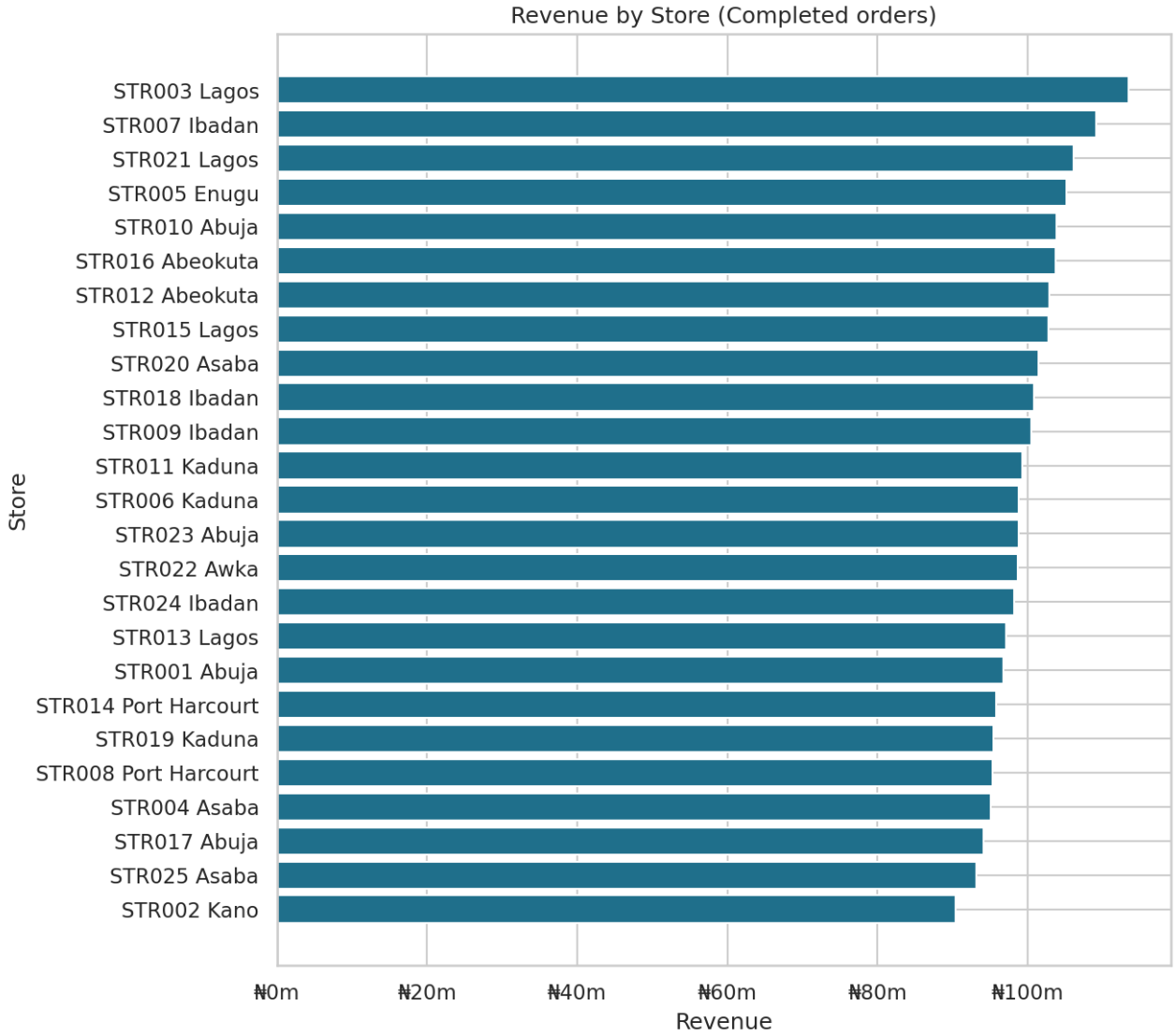


Figure 3 — Revenue by store. Best store STR003 Lagos, weakest STR002 Kano (+25.5% on Completed orders; only +6.0% on all orders).

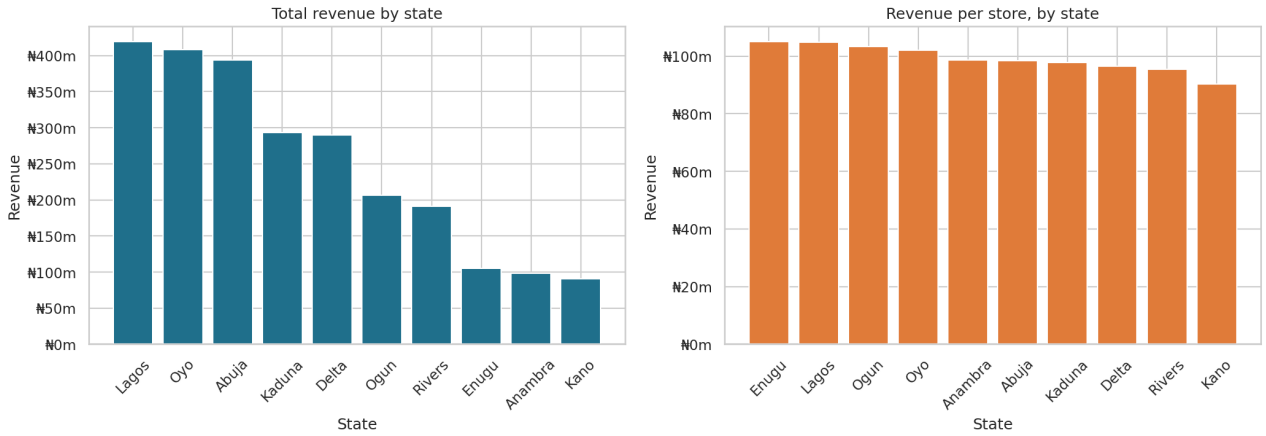
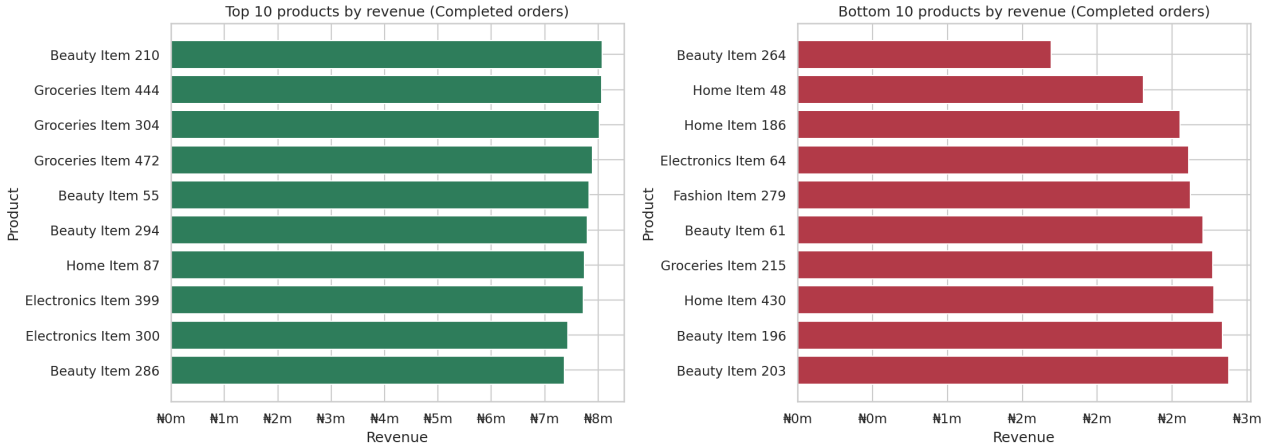


Figure 4 — Revenue by state. Total revenue follows the number of stores; per store the states are almost identical.



Figures 5 and 6 — Top 10 and bottom 10 products. The gap between best and worst is small; the lists change when all orders are counted.

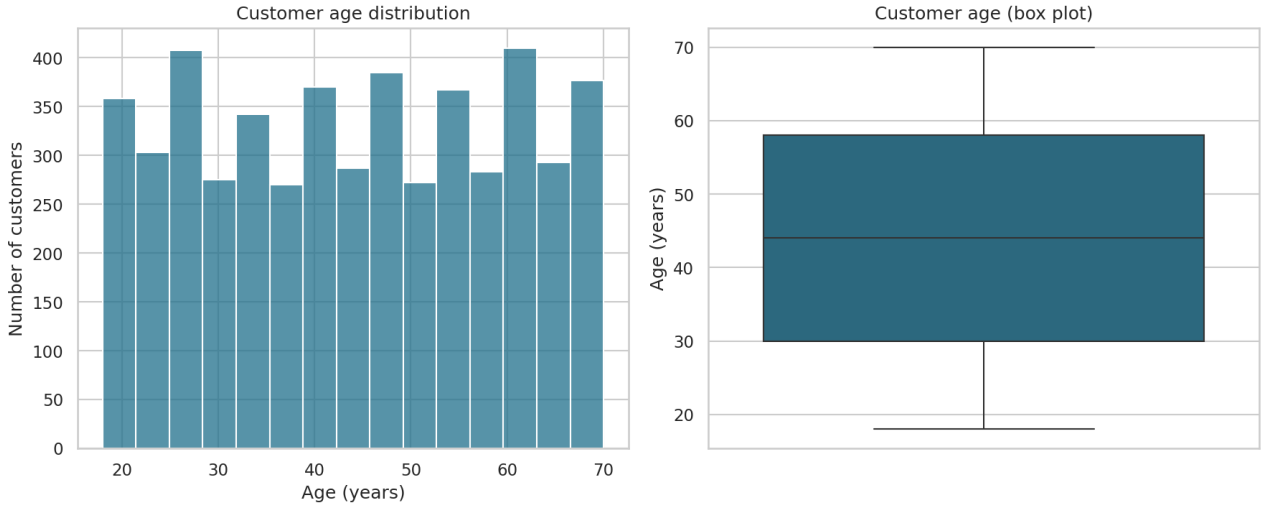


Figure 7 — Customer age. Customers are spread evenly from 18 to 70 (mean 44).

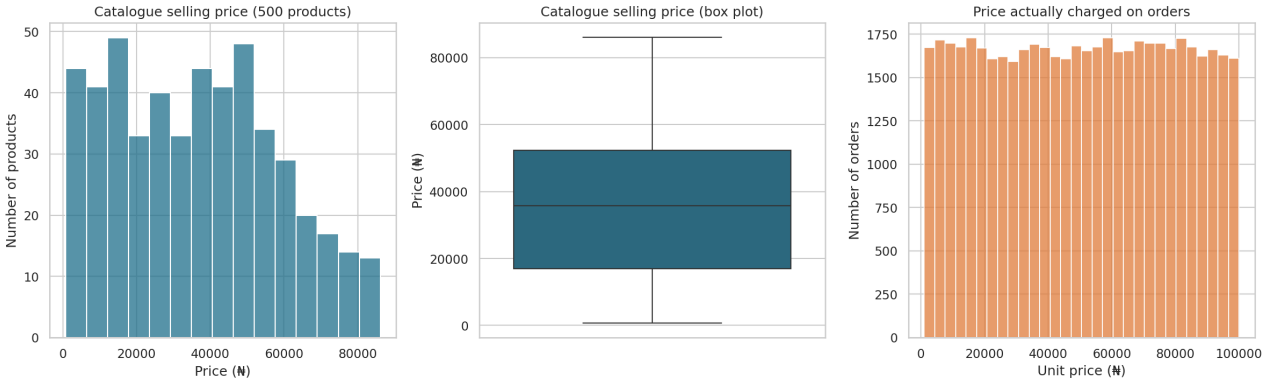


Figure 8 — Product prices. Catalogue prices (left, middle) run ₦676–₦86,153; prices charged on orders (right) are flat from ₦1,003 to ₦99,998.

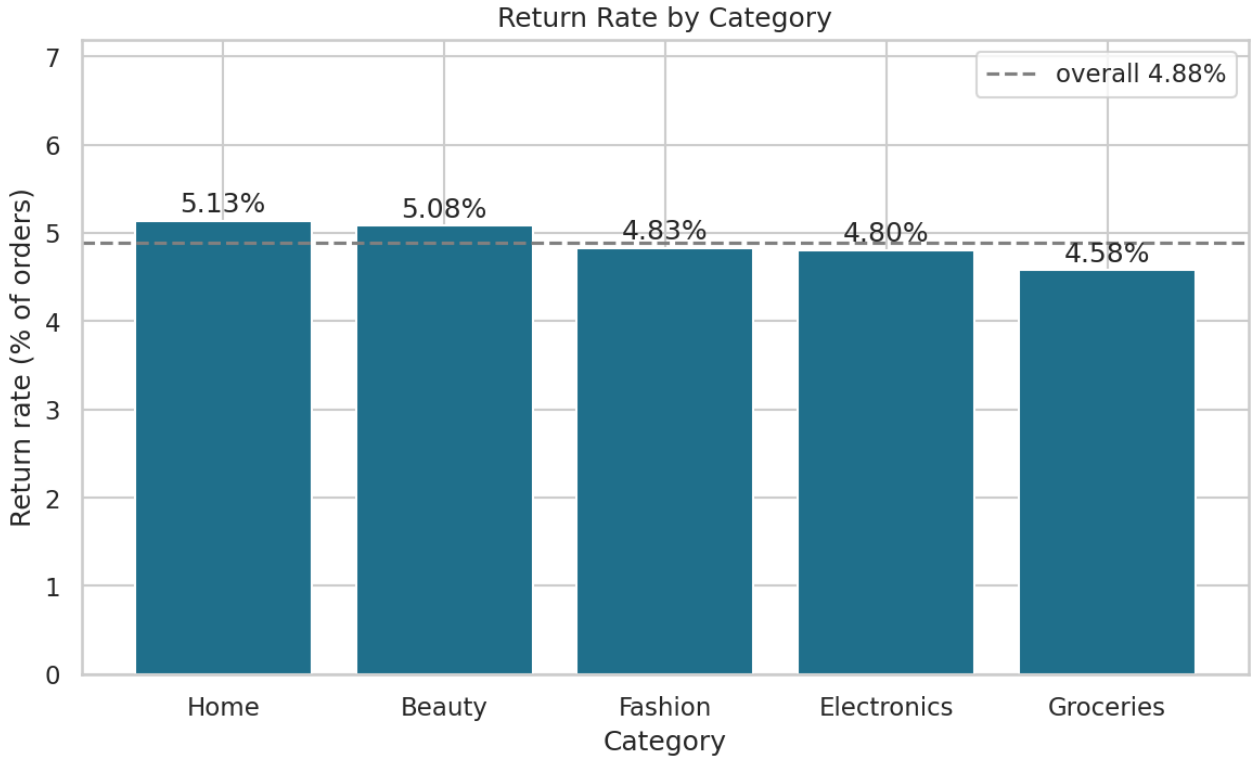


Figure 9 — Return rate by category. All categories sit close to the overall 4.88%.

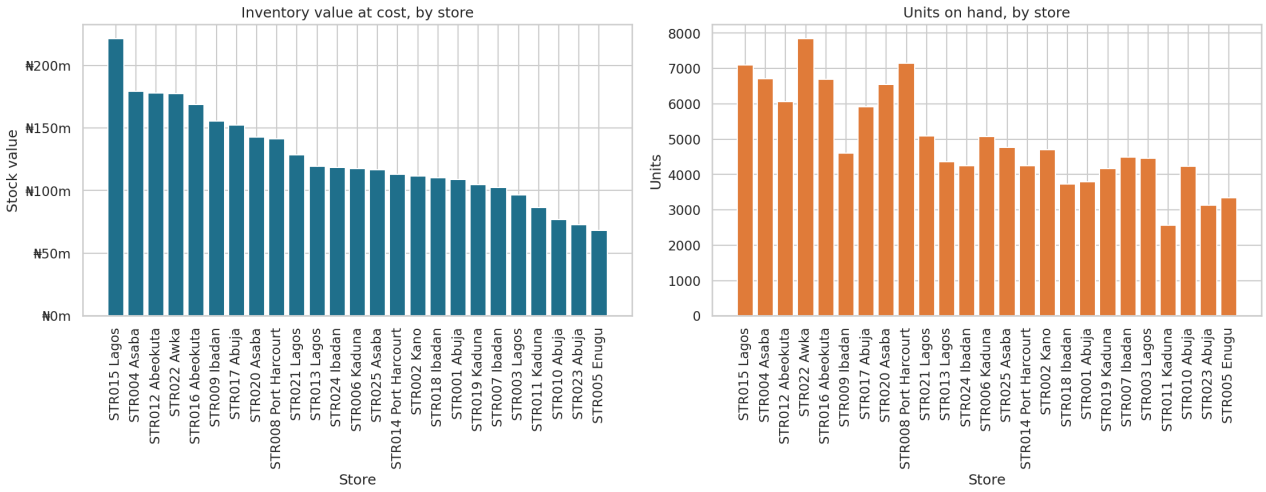


Figure 10 — Inventory by store. Value follows the number of products held (9–27 per store); each product sits in one store only.

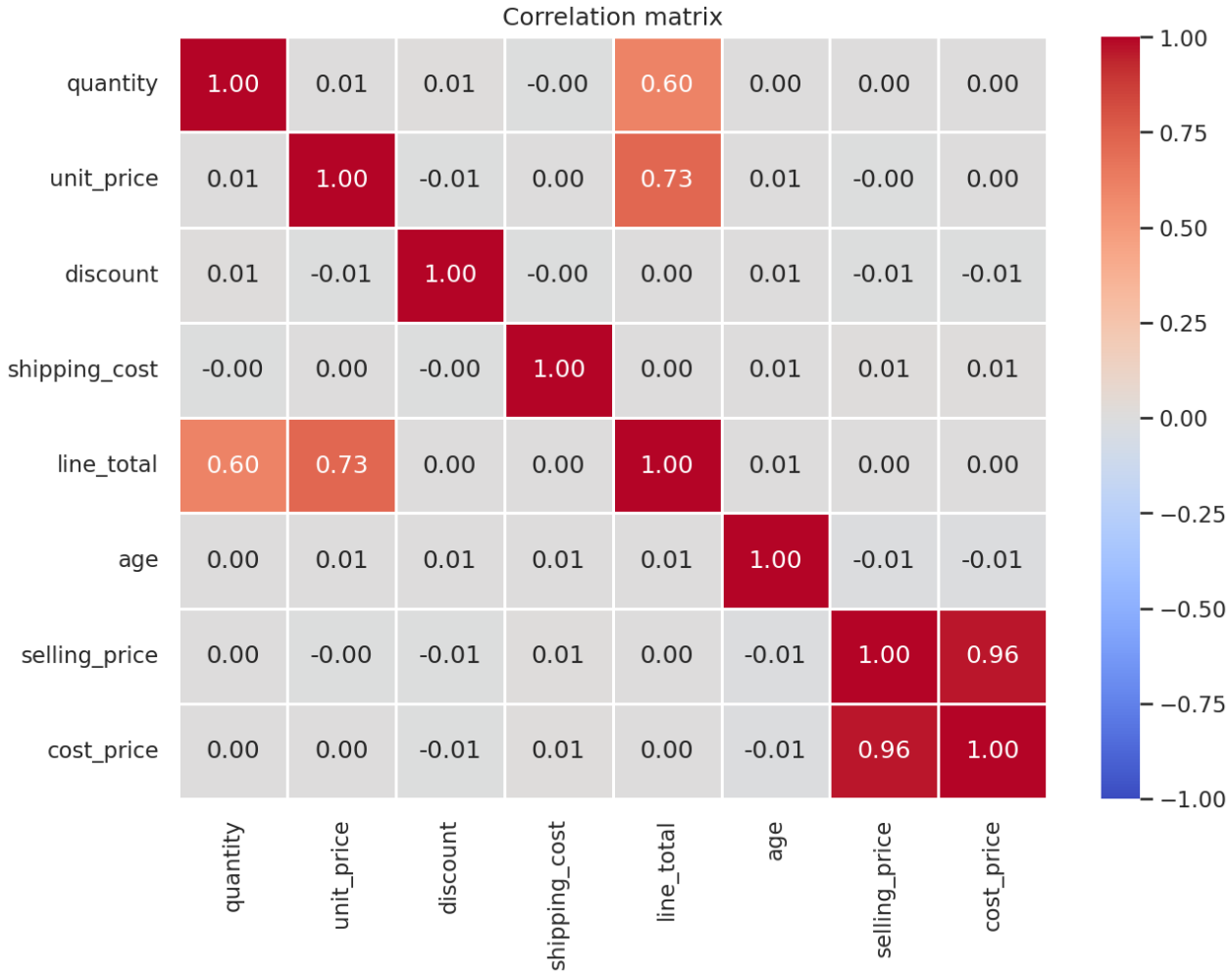


Figure 11 — Correlation matrix. Only formula-based links and the cost/price link stand out.

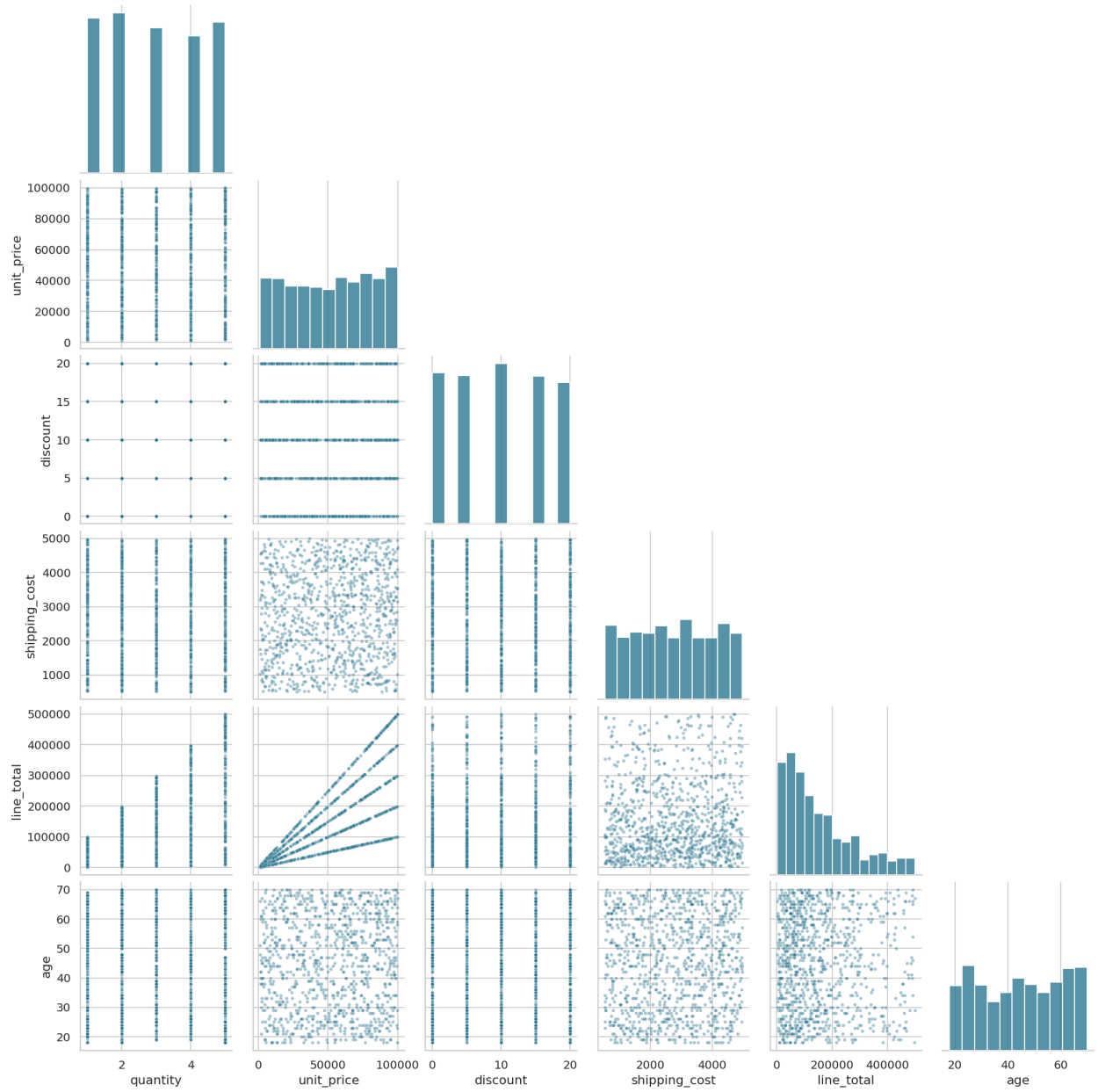


Figure 12 — Pair plot (1,000-order sample). Structure appears only where a formula links the variables.

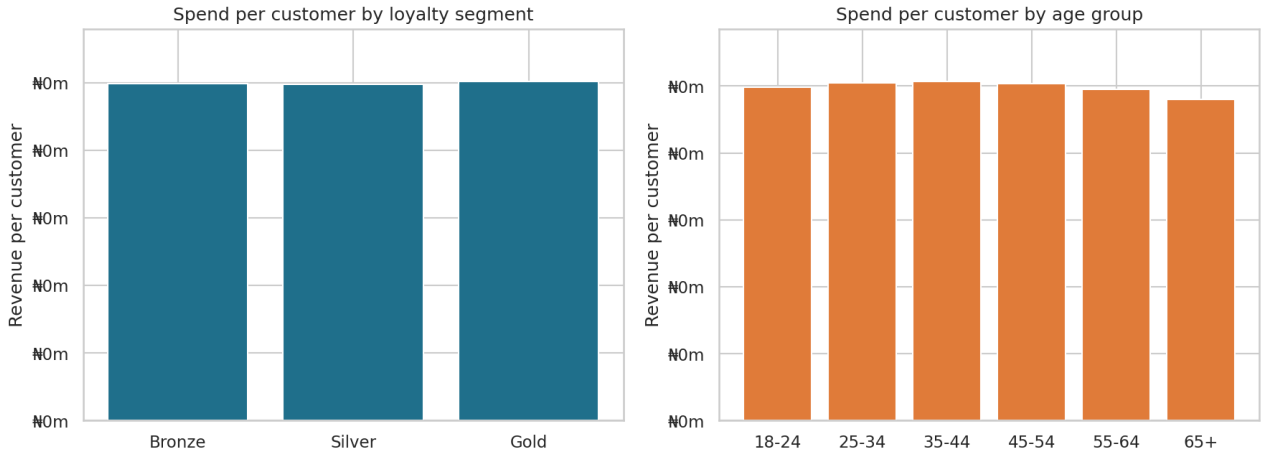


Figure 13 — Spend per customer by loyalty segment and age group: almost identical.

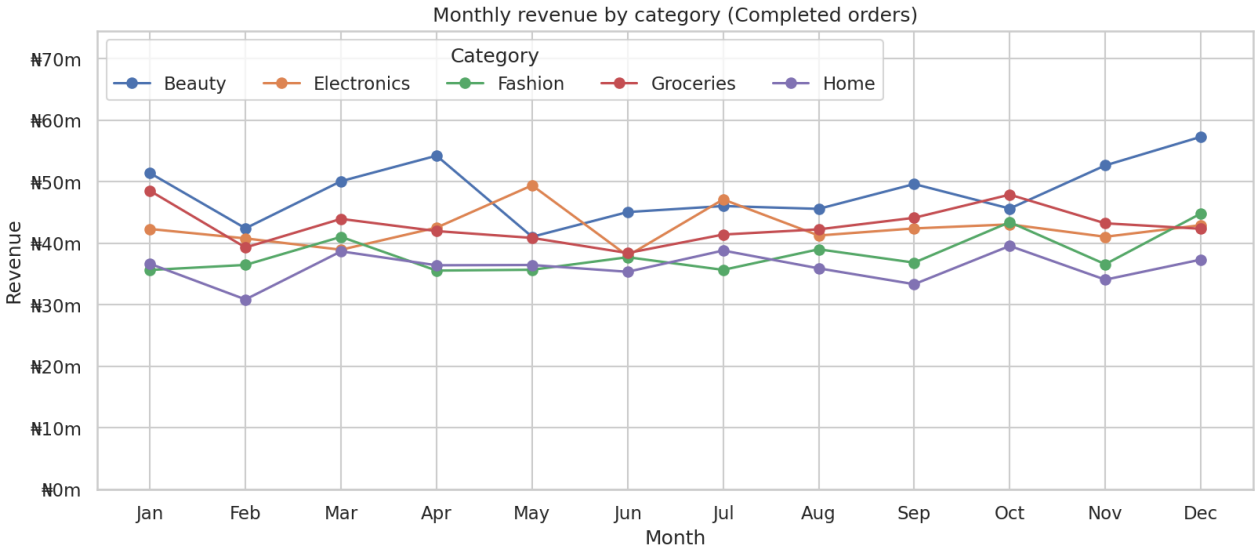


Figure 14 — Monthly revenue by category. Lines move up and down but show no clear growth.

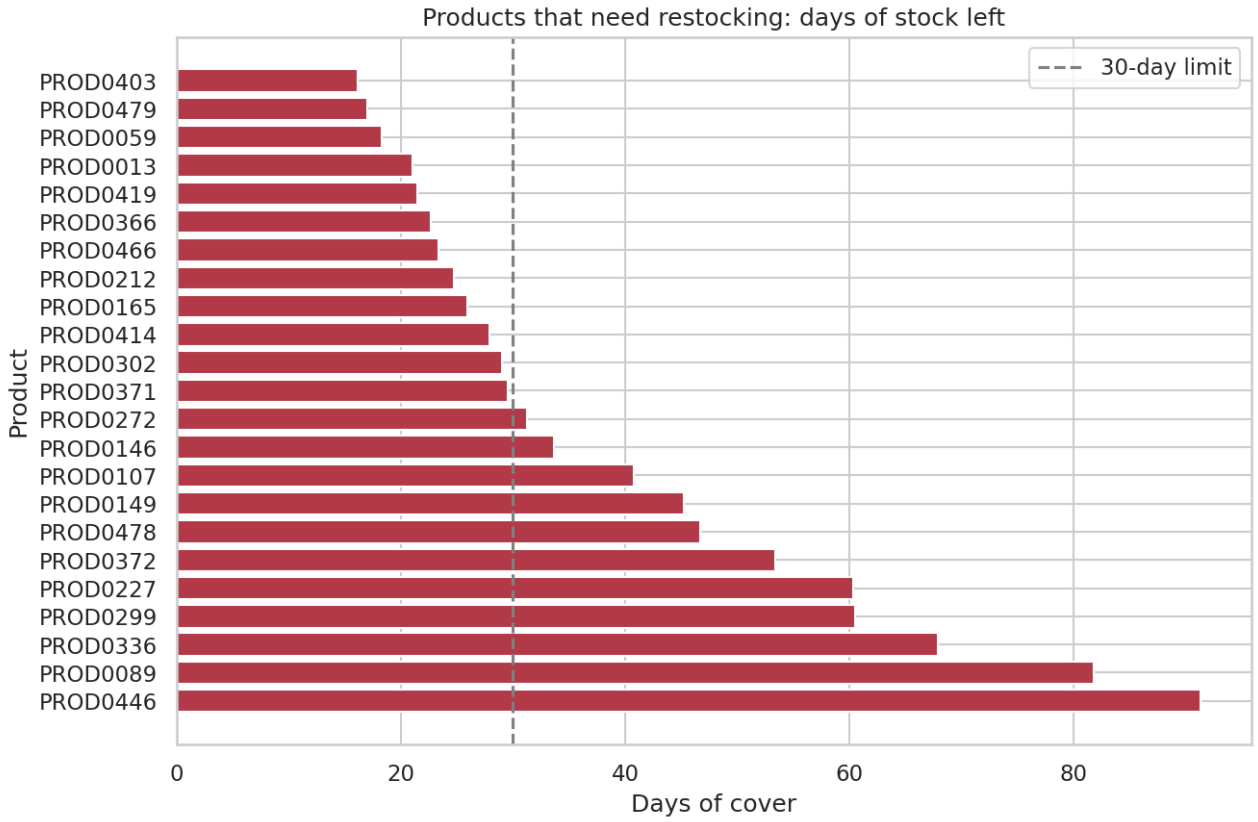


Figure 15 — The 23 products that need restocking, by days of stock left.



Figure 16 — Revenue per store by region: gaps are small.

6. Five Key Insights

Insight 1: Only a third of booked value is Completed

₦2.50bn (33%) is Completed, ₦2.55bn (34%) is Cancelled and ₦2.53bn (33%) is Pending. The three statuses are almost equal and unrelated to channel, payment method or category.

Insight 2: Stores, regions, segments, categories and months perform alike

The store gap is 6.0% on all orders (and the best store changes when only Completed orders are counted); Gold customers spend the same as Bronze; February is the only low month and it is shorter. Differences are within normal ups and downs.

Insight 3: Pricing data is unreliable

24% of orders were sold below cost price, the price charged is unrelated to the price list, and the discount is applied as a flat naira amount. The median profit margin is 50% but 24% of orders have a negative margin.

Insight 4: Return records fail logic checks

50% of return dates come before the order, 67% belong to orders that were never completed, 270 refunds exceed the sale, and the return rate is about 5% in every product category and customer group.

Insight 5: Inventory is unbalanced

23 products need restocking now (about ₦7.9m for 60 days of cover) while 307 products hold over a year of stock worth ₦2.72bn, 86% of inventory value.

7. Five Recommendations

#	Recommendation	Owner
1	**Reduce cancellations and pending orders.** Find the causes of the one-third cancelled and one-third pending orders, set a maximum pending time, and report Completed revenue as the main KPI.	Operations
2	**Govern prices and discounts.** Confirm whether Discount is a percentage or a naira amount and restate revenue; make till prices follow the price list; review every below-cost sale.	Commercial / Finance
3	**Add rules to the returns process.** Return date after order date; refund not above the sale; only Completed orders can be returned; record the reason and product quality.	Customer Service / IT
4	**Rebalance inventory.** Restock the 23 urgent products and set a maximum stock level so money is not tied up in over a year of stock.	Supply Chain
5	**Build the data foundation.** Write a real data dictionary, redefine regions geographically, record stock per store, delivery time and product quality, and collect 2–3 years of history.	Data / IT

8. Suggested Machine Learning Problems

Feasibility is judged from what this analysis found. Return rates are the same in every group and sales are flat, so most prediction problems need better data first.

ML problem	Possible now?	What is needed
Detecting unusual transactions (below-cost sales, refunds above the sale)	**Yes**	Rules or simple statistics; nothing more
Customer segmentation (K-Means on recency, frequency, spend)	**Partly**	More behaviour data for meaningful groups
Inventory forecasting / planning (days of cover)	**Yes, as rules**	Stock per store; longer sales history
Sales forecasting	**Not yet**	2–3 years of history
Return prediction	**No**	Product quality, delivery and customer-history data; correct return dates
Customer churn prediction / product recommendation	**No**	Multi-year history; multi-item baskets

9. Assumptions and Limitations

Assumptions

- **Currency** is Nigerian naira (₦), inferred from store locations.
- **Revenue** = Line_Total on Completed orders. All-orders figures are shown where stated.
- **Discount** is kept as recorded (a flat amount); whether it is a percentage is a question for the client.
- **Gross profit** uses the catalogue cost price, so margins are indicative only.
- **Customer lifetime value** = profit in the year × an assumed 3-year lifespan (an approximation for ranking customers).
- **Days since last purchase** is measured from 1 January 2026 (the day after the last order).
- **Inventory status:** Low = at or below the reorder level or under 30 days of stock; High = more than 365 days; calculated from system-wide sales because each product is stocked in one store.
- **Suppliers** are linked to products through the brand name; no delivery or quality data exist.

Limitations

- Only one year of data, so no real trend or seasonality can be measured.
- With about 660 completed orders per store and 100 orders per product, small differences are partly luck; where a ranking changed with the base used, this report says so.
- Several fields (return dates, refunds, registration dates, prices) are inconsistent, so results that depend on them are indicative.
- The optional bonus challenge was not attempted in this submission.

Reproducing this work

Install *requirements.txt*, open *Week1_EDA.ipynb* from the project folder and choose *Restart & Run All*. The notebook reads *Raw_Data/* and writes *Cleaned_Data/* and *Visualizations/*.